CHAPTER 4

LAWS OF MOTION

CHAPTER ESSENCE AT A GLANCE:
Things keep doing what they are doing unless a push or pull interferes! Newton's 2nd Law (F = dp divided by dt = ma) is the heart of classical physics. Banking curved roads prevents vehicles from skidding because normal force lends a helping hand towards the center!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Inertia & Momentum:** Inertia is the natural tendency to resist change in motion (measured by mass m). Momentum p = mv is a vector having the direction of velocity.

• **Newton's 2nd Law & Impulse:** F = dp divided by dt = ma. Impulse I = F · Δt = Δp = m(v - u) equals the change in momentum (also given by area under Force-Time graph).

• **Friction Laws:** Static friction f s ≤ μ s N self-adjusts to match applied force until limiting friction. Kinetic friction f k = μ k N opposes sliding (μ k < μ s). Friction is independent of contact area!

• **Circular Turning & Banking:** Level road safe speed: v max = sqrt(μ s R g) (cyclist slips if speed exceeds this!). Banked road optimum speed: v_0 = sqrt(R g tan θ). With friction: v max = sqrt(R g (μ s + tan θ) divided by (1 - μ s tan θ)).

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = m\mathbf{a}$$

Newton's Second Law of Motion

$$\mathbf{I} = \mathbf{F}\Delta t = \Delta\mathbf{p} = m(\mathbf{v} - \mathbf{u})$$

Impulse-Momentum Theorem

$$f_s \leq \mu_s N, \quad f_k = \mu_k N, \quad \tan\theta = \mu_s$$

Laws of Friction and Angle of Repose

$$v_{\max} = \sqrt{Rg\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}} \quad \left(v_0 = \sqrt{Rg\tan\theta}\right)$$

Maximum Safe Speed on Banked Circular Road

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Normal force on an inclined plane is N = mg cos θ, NOT mg! Friction is along the plane: f = μ N = μ mg cos θ.

• **Alert:** Momentum is always conserved during collisions whether kinetic energy is conserved or not!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State Newton's second law of motion and express it in vector mathematical form. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Statement]** The rate of change of linear momentum of a body is directly proportional to the applied net external force and takes place in the direction of the force. [1 Mark]

• **[Step 2: Vector Equation]** F ext = dp divided by dt = d(m v) divided by dt = m (dv divided by dt) = m a. [1 Mark]

• **Key Exam Result:** F = dp divided by dt = m a

2. Explain why a cricket player lowers his hands backward while catching a fast cricket ball. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Impulse Principle]** From impulse-momentum theorem: Impulse = F x Δt = Δp (fixed momentum change). Force F = Δp divided by Δt. [1 Mark]

• **[Step 2: Explanation]** By drawing hands backward, the fielder increases the time interval Δt of the catch. Since force is inversely proportional to time (F proportional to 1 divided by Δt), increasing Δt greatly reduces the impact force on the player's hands, preventing injury. [1 Mark]

• **Key Exam Result:** Increasing contact time Δt reduces impact force F

3. Show that Newton's second law is the real law of motion (i.e. it contains both the first and third laws). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Contains First Law]** From second law: F = m a. If no external force acts (F = 0): m a = 0 => a = 0 (since m $\neq$ 0). Acceleration a = 0 implies velocity remains constant (v = const), meaning a body at rest stays at rest and a moving body continues moving uniformly in a straight line, which is the first law. [1.5 Marks]

• **[Step 2: Contains Third Law]** Consider an isolated two-particle system with no external force. Total momentum is conserved: dp total divided by dt = 0 => dp1 divided by dt + dp2 divided by dt = 0 => F12 + F21 = 0 => F12 = -F21 (Action = -Reaction), which is the third law. [1.5 Marks]

• **Key Exam Result:** Second law contains both first and third laws

4. State the law of conservation of linear momentum. Derive the expression for the recoil velocity of a gun of mass M firing a bullet of mass m with velocity v. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Statement]** The total linear momentum of an isolated system remains constant in the absence of any net external force (F ext = 0 => p = constant). [1 Mark]

• **[Step 2: Derivation of Recoil]** Initial momentum before firing (both at rest): p initial = 0. After firing, let V be the recoil velocity of gun and v be bullet velocity: p final = M V + m v. By conservation of momentum: p final = p initial => M V + m v = 0 => M V = -m v => V = -(m divided by M) v. The negative sign shows recoil is opposite to bullet motion. [2 Marks]

• **Key Exam Result:** Recoil Velocity V = -(m divided by M) v

5. Differentiate between static friction, limiting friction, and kinetic friction. State the laws of limiting friction. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Types of Friction]** 1. Static friction (f s): Opposes impending motion; self-adjusting from 0 up to a maximum. 2. Limiting friction (f L): Maximum static friction just before sliding begins (f L = μ s N). 3. Kinetic friction (f k): Friction during actual sliding (f k = μ k N, with f k < f L). [1.5 Marks]

• **[Step 2: Laws of Limiting Friction]** (i) Limiting friction opposes direction of impending motion. (ii) It is directly proportional to normal reaction: f L = μ s N. (iii) It is independent of the apparent area of contact. (iv) It depends on the nature and roughness of materials in contact. [1.5 Marks]

• **Key Exam Result:** f L = μ s N, f k = μ k N (f k < f L)

6. A bullet of mass 0.04 kg moving with a speed of 90 m/s enters a heavy wooden block and is stopped after penetrating 60 cm. Calculate the average resistive force exerted by the block. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Given Data]** Mass m = 0.04 kg, initial speed u = 90 m/s, final speed v = 0, distance s = 60 cm = 0.6 m. [0.5 Mark]

• **[Step 2: Retardation Calculation]** From $v^2 = u^2 + 2as$: $0 = 90^2 + 2a(0.6) \Rightarrow 1.2a = -8100 \Rightarrow a = -8100 / 1.2 = -6750$ m/s^2. [1.5 Marks]

• **[Step 3: Average Resistive Force]** F = m a = 0.04 x (-6750) = -270 N. Magnitude of resistive force = 270 N. [1 Mark]

• **Key Exam Result:** Average Resistive Force = 270 N

7. A body of mass 5 kg is acted upon by two perpendicular forces 8 N and 6 N. Find the magnitude and direction of the acceleration of the body. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Resultant Force]** Since forces F1 = 8 N and F2 = 6 N are perpendicular ($\theta = 90°$): $F_{net} = \sqrt{(F1^2 + F2^2)} = \sqrt{(8^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{(100)} = 10$ N. [1 Mark]

• **[Step 2: Acceleration Magnitude]** a = F net / m = 10 N / 5 kg = 2 m/s^2. [1 Mark]

• **[Step 3: Direction of Acceleration]** $\tan\theta = F2 / F1 = 6 / 8 = 0.75 \Rightarrow \theta = \tan^{-1}(0.75) \approx 36.87°$ with the 8 N force. [1 Mark]

• **Key Exam Result:** a = 2 m/s^2 at angle 36.9° to the 8 N force